Preston Crowe

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Security Engineer with 5+ years experience in detection engineering, threat intelligence infrastructure, and vulnerability research.

[EXPERIENCE]

ExtraHop — Seattle, WA
Security Engineer III, December 2020 – Present

- Shipped detections within hours of disclosure for critical vulnerabilities (e.g., Log4Shell, SUPERNOVA) and partnered with technical writers to publish expedited threat briefings during incident response.
- Built an Apache Iceberg indicator data lake serving analyst queries at 100M+ indicator scale; exported collections to a production enrichment API powering detections for all customers.
- Built the first ExtraHop MCP server for agentic enumeration of detections and devices, winning the 2025 hackathon audience choice award.
- Automated AWS traffic-mirroring environments to correlate detections against breach and attack simulation platforms; simulated 3,000+ Mandiant Security Validation (MSV) file transfer actions and added their indicators to the detection catalog, covering over half of MSV's available actions.
- Authored the internal MSV testing guide and worked with sales engineers on best-practice configurations in customer engagements.
- Developed detection strategies for red team tools (Redline, BloodHound, PetitPotam, Farmer); built pcap manipulation, device discovery, and automated traffic replay tooling with Threat Research and performance teams.
- Automated CVE monitoring and Jira ticket creation/triage for network vulnerabilities, eliminating a weekly one-hour meeting for the ~10-person threat research team (520 researcher-hours per year); published daily CVE summaries and sentiment analysis.

Baffin Bay Networks — Seattle, WA
Internship, Product Development – June 2019 – September 2019

- Built an internal dashboard to monitor a honeypot network in React. Aggregated health metrics for sensors hosted across dozens of cloud providers worldwide.
- Developed an API for Baffin Bay's Threat Insight platform to provide IP context enrichment.

F5 Networks — Seattle, WA
Internship, F5 Labs – June 2018 – September 2018

- Worked with a team of security researchers to develop a vulnerability data pipeline to monitor classes of CVEs.
- Built a threat map with React and D3.js to visualize suspicious traffic in real time.
- Generated data visualizations and animations for a Black Hat 2018 talk.

NASA AMES Research Center — Mountain View, CA
Internship, OpenMCT – June 2017 – September 2017

- Implemented JSON import/export functionality for workspace hierarchies in an open-source mission control application.

[SKILLS]

- **Detection & Response:** detection engineering, detection validation, threat briefings, vulnerability triage, incident response
- **Threat Intelligence:** indicator sourcing and enrichment, TI collections, IDS, CVE and exploit monitoring, OSINT
- **Adversary Tradecraft:** exploit reproduction, honeypots, red team tool analysis
- **Network Security:** PCAP analysis and generation, traffic mirroring and replay, device discovery
- **Data & AI Engineering:** Apache Iceberg,ETL, SQL, enrichment APIs, MCP, agentic tooling, data lakes
- **Languages & Frameworks:** Python, Go, JavaScript, Java, C, React, Vue.js, D3.js
- **Cloud & Infrastructure:** AWS, Linux, Docker, Terraform, Ansible, GitLab CI/CD
- **Interests:** Proxmox, OPNSense, OpenWrt, SDR, ESP32, LoRa, 3D printing, data hoarding

[EDUCATION]

University of Washington — Seattle, WA
Sep 2016 – Jun 2020
B.S. Computer Science, Dean's List, 3.71 GPA

[PUBLICATIONS]

- F5 Networks, 2018: *Breaking Down the Door to Emergency Services through Cellular IoT Gateways*
- ExtraHop, 2021: *PetitPotam: Expanding NTLM Relay Attacks*